Report on "Matrix Theory Reloaded: A BPS Road to Holography" (Blair, Lahnsteiner, Obers, Yan)

Summary

The paper revisits the decoupling limits that define matrix theories on Dp-branes (matrix p-brane theory, MpT), stressing that they are BPS limits acting on the whole string theory. The resulting ten-dimensional target space is non-Lorentzian, with a foliation adapted to the chosen brane. Applied to curved D-brane backgrounds, a single such limit at asymptotic infinity corresponds to the near-horizon limit in the bulk. Two successive limits give further holographic examples, including ones with non-Lorentzian bulk. Uplifted to M-theory, each limit is a DLCQ, which motivates the conjecture that holography is a DLCQ^n/DLCQ^m correspondence with $m > n$. Undoing the limits is a deformation of the matrix theory. This is related to the $T\bar{T}$ deformation, and it yields $T\bar{T}$-like flow equations for the Dp-brane DBI action.

1 Principal achievement

The main achievement is a unifying, systematic framework. It links matrix theory, AdS/CFT, non-Lorentzian geometry and $T\bar{T}$ deformations through one organising principle, the BPS decoupling limit. It answers a long-standing puzzle: how asymptotically non-Lorentzian matrix-theory geometry relates to the Lorentzian AdS bulk. The answer is that the near-horizon limit is the extra DLCQ. The same logic reproduces AdS_5, the D1-D5 AdS_3 case (as DLCQ^0/DLCQ^2), and the M2 and M5 cases. It also gives a mechanism for generating new, non-Lorentzian holographic pairs. The explicit p-brane $T\bar{T}$ flow equations, such as the strikingly simple three-dimensional M2/D2 flow, are concrete and checkable results. They allow the bulk geometry to be built intrinsically, without taking a limit.

2 Principal drawback

The central claims are largely structural and conjectural, and they are not tested dynamically. The DLCQ^n/DLCQ^m proposal is supported by examples and pattern-matching, not by derivation or by a precise operator or observable map. The new non-Lorentzian bulk geometries are obtained by hand-inserting a rescaling of the harmonic function. The authors admit they have not checked that these geometries solve the equations of motion and torsional constraints of MpT supergravity, which has not been formulated. They also leave open the isometries, the physical interpretation of the N rescaling, and the choice between smeared and localised solutions. The flow equations are bosonic and abelian only, so the connection to actual matrix theory (non-abelian SYM) is untested. Results for $p > 3$ rely on strongly coupled descriptions, and the non-Lorentzian description at infinity may be coordinate-dependent and tied to the Poincaré patch.

3 Most promising follow-up idea

Balancing feasibility and impact, I recommend **formulating MpT supergravity (at least bosonically) and using it to verify the proposed non-Lorentzian near-horizon solutions**. The first target would be the M2/M0T/M1T-type backgrounds that already have partial results in non-relativistic M-theory. This would compute their isometry algebras and supersymmetries, and it would test the smeared versus localised choice. It is feasible because the non-Lorentzian supergravity and Newton–Cartan technology, the algebras (the M0T superalgebra) and the explicit candidate solutions already exist, and a limit of the Lorentzian equations is a well-defined calculation. The payoff is high. It would turn the DLCQ^n/DLCQ^m picture from a conjecture into verified non-Lorentzian holography, and it would fix the gravity side for the second step, which is to test the flow equations. That second step is to extend the $T\bar{T}$-like flows to supersymmetric or non-abelian cases, such as the M2 flow applied to BLG/ABJM, and to compare observables such as deformed spectra with the corresponding bulk solutions. The more ambitious ideas (Carrollian bulks from MpT with $p < -1$, matching finite-N BFSS amplitudes) have higher potential impact but are much less tractable.